

Dead Man's Cipher — ELI5 Guide (Hinglish)

Hinglish Edition • Easy Hindi + English Explanation

DEAD MAN'S CIPHER
Hackathon Spec Document
ELI5 Edition

1. Executive Summary (Hinglish ELI5 Concept)

Dead Man's Cipher ek pirate-themed cryptographic workbench hai jo secret messages ko encrypt, digitally sign, lyrics ya images me hide, aur inspect karne ke liye banaya gaya hai. Yeh pirates ko dushman ships aur codebreakers se bachakar safe coordinates bhejne me madad karta hai.

2. Tech Stack & Tools List (Aasan Bhasha Mein)

Tool / Library	Module Name	Kaam & Purpose (Hinglish)
Next.js 16	App Router & Turbopack	Fast React web app framework for layout & routing.
TypeScript	Strict Type Checks	Code me zero runtime type errors guarantee karta hai.
WebCrypto API	window.crypto.subtle	Browser ka built-in military encryption: Ed25519, AES-256-GCM & PBKDF2.
Steganography	Zero-Width Unicode & Canvas 2D	Text lyrics me (u200B) aur PNG images ke pixels me data chupata hai.
Web Audio API	AudioContext Synthesizer	Pure JS se ship bell, clicks, alarm sirens aur chimes bajata hai.
Recharts	BarChart & RadarChart	Live frequency charts aur 5-axis security radar chart draw karta hai.
Tailwind CSS v4	Custom Design System	Responsive dark theme aur spacious clean UI layout formatting.

3. Feature-by-Feature Deep Dive & Logic (Hinglish)

Har ek feature kaise kaam karta hai uska complete step-by-step logic:

Feature A: Identity Vault (/identity)

- **WebCrypto Keypairs:** Ed25519 private/public keys generate karta hai.
- **8-Character Fingerprint:** Public key ka SHA-256 hash nikal kar 8-digit unique ID (jaise 8F3A2B1C) banata hai.
- **Printable SVG Wax Seal:** High-resolution vector SVG seal draw karta hai jisme Captain name, Ship aur fingerprint hota hai.
- **BLACKSPOT Key Revocation:** Agar key chori ho jaye toh use BLACKSPOT mark kar sakte ho. Inspection me BLACKSPOT key wala message turant block ho jata hai.

Feature B: Sealed Missive Forge (/compose)

- **PBKDF2 Key Derivation:** Passphrase se 100,000 rounds PBKDF2-SHA256 chalakar 256-bit encryption key banata hai.
- **AES-256-GCM Encryption:** Message ko random IV aur GCM authentication tag se lock karta hai.
- **Ed25519 Digital Signature:** Message ke upar Captain ka signature lagata hai taaki authenticity prove ho sake.
- **Roll Random Missive:** 1-click me random pirate location, passphrase, identity aur shanty generate karta hai.

Feature C: Boarding Inspection (/inspect)

- **4-Gate Defense Pipeline:** Incoming messages ko 4 security gates se verify karta hai:
 1. Gate 1 (Format): Protocol header DMC1 verify karta hai.
 2. Gate 2 (Replay Window): 24 ghante se purane messages ko reject karta hai.
 3. Gate 3 (Signature): Ed25519 signature check karta hai.
 4. Gate 4 (Revocation): BLACKSPOT keys ko block karta hai.
- **Audio Feedback:** Gate pass hone par ship bell aur fail hone par alarm siren bajti hai.

Feature D: Steganography Lab (/stego)

- **Sea Shanty Text Stego:** Secret text ko binary (0 aur 1) me convert karke invisible Unicode characters (\u200B & \u200C) banakar lyrics me chupa deta hai.
- **Sail Emblem Image LSB Stego:** HTML5 Canvas API se PNG image pixels read karta hai aur RGB color bytes ke Least Significant Bit (LSB) me secret location store karta hai.
- **Interactive Map Picker:** Direct NauticalMap se location select karke image stego me hide kar sakte ho.

Feature E: Kraken's Trial Attack Lab (/attack-lab)

- **4 Simulated Attacks:** Forged Signatures, 72-Hour Replay Attacks, BLACKSPOT Keys, aur Tampered Ciphertexts.
- **Defense Verification:** Prove karta hai ki system ke 4 gates har cyber attack ko pakad kar reject kar dete hai.

Feature F: Cipher Cracker & Encrypter (/cracker)

- **Cryptanalysis Engine:** Letter frequency histograms, Index of Coincidence (IoC ~0.0667 English vs ~0.038 random), aur Chi-Squared Caesar solvers dikhata hai.
- **Spacious Interactive Encrypter:** Bada plaintext textarea, key shift slider (+1 to +25), aur 1-click "Encrypt & Load into Codebreaker" button presets ke sath offer karta hai.

Feature G: Threat Dashboard & Ship's Log Audit

- **Threat Dashboard (/threat):** 5-axis Radar Chart (Cipher Strength, Key Freshness, Sig Assurance, Anti-Replay, Stego Stealth) aur 96/100 score gauge.
- **Ship's Log Audit (/audit):** Tamper-evident SHA-256 hash-chained log (currentHash = SHA256(previousHash : seq : timestamp : event : actor : details)). 1-click integrity check se tampering pakdi jati hai.